

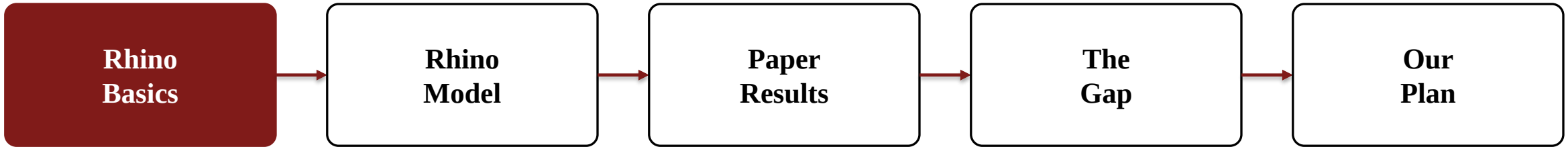
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Date: 12th November 2026
Program: B.Tech CSE
Domain: Rhino, deep temporal causal discovery

Can a Deep Temporal Causal Posterior Be Trusted?

What This Talk Covers

Five parts, from the base paper to our own plan.



What We Want To Learn

Who drives whom, and after how long.

The data are several variables over time. The result is a graph.

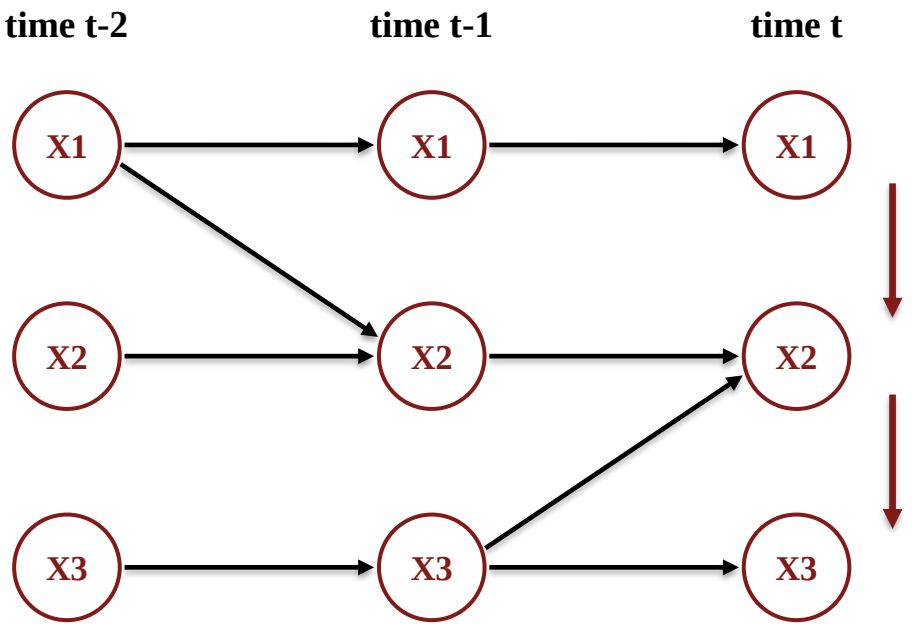


Fig. 1: Black, lagged edges. Maroon, instantaneous edges (no cycle allowed).

Why It Is Hard

Real series have three awkward features.

Older methods handle one or two of them.
Rhino aims at all three.

Non-linear effects
An announcement acts on a price in a curved way.

Instantaneous effects
Slow sampling folds a day of events into one step.

History-dependent noise
A student with recent good marks scores with little variation.

What Older Methods Miss

No earlier family covers all three features.

Rhino is built to close this niche, with a proof that the graph is identifiable.

Family	Examples	What it misses
Granger causality (incl. deep)	cMLP, cLSTM, TCDF, SRU	No instantaneous effects, fixed noise
Functional models	VAR-LiNGAM, DYNOTEARS	Linear, fixed noise
Constraint based	PCMCI+	Equivalence class only, slow for many nodes
Rhino	this paper	Non-linear, instantaneous, history-dependent noise

Rhino In One Equation

A mean part plus a noise part.

The graph decides which parents each part may use.

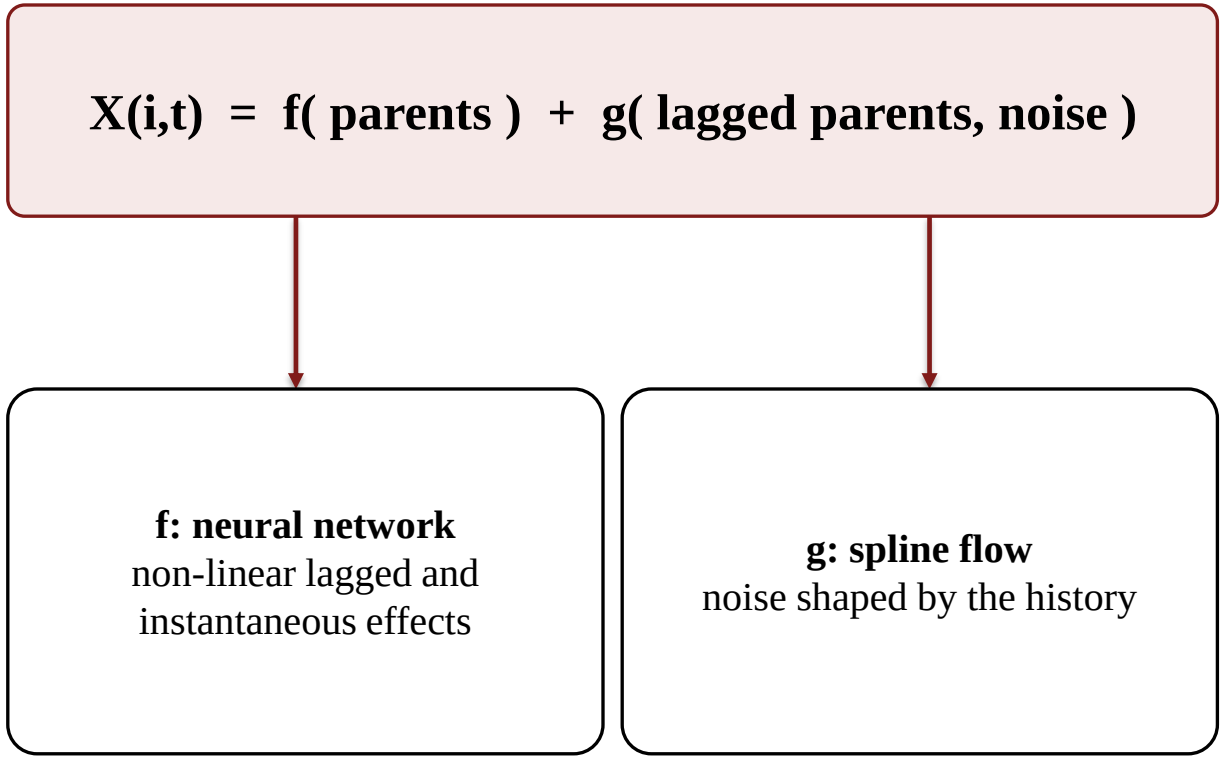


Fig. 2: Rhino's structural equation model.

History-Dependent Noise

g lets the noise change with the past.

On the right the spread grows with the past value. A fixed-noise model cannot do this.

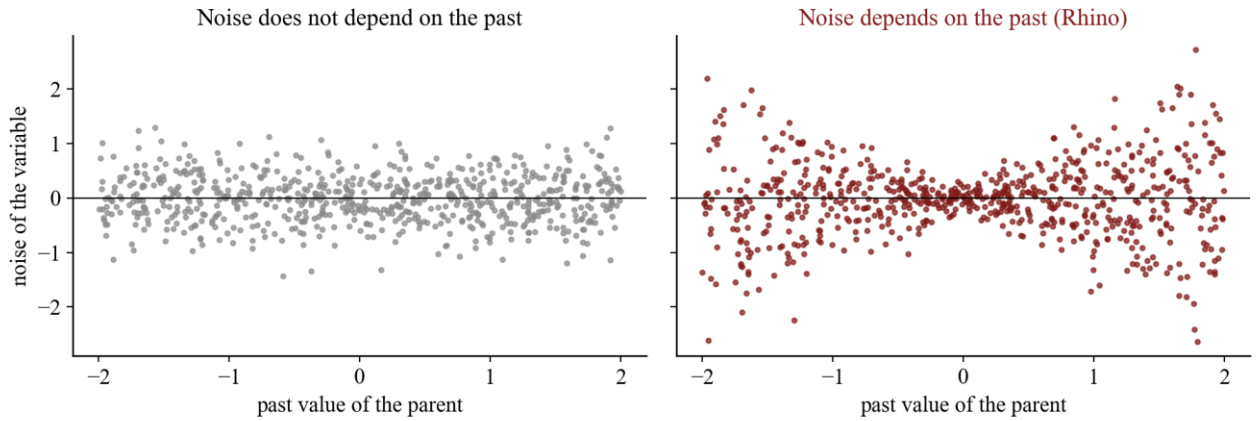


Fig. 3: Simulated illustration of fixed and history-dependent noise.

The Graph Is A Set Of Switches

A filled cell turns one parent on.

Learning the graph means learning which cells are on.

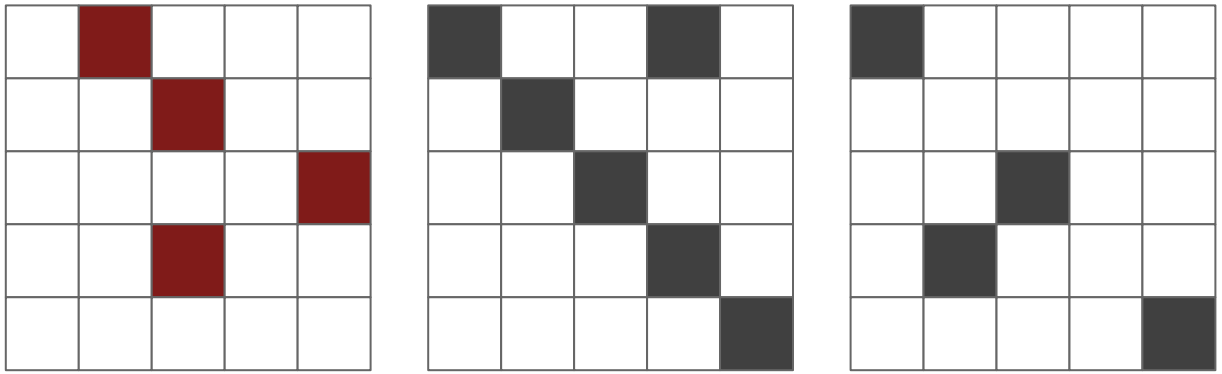
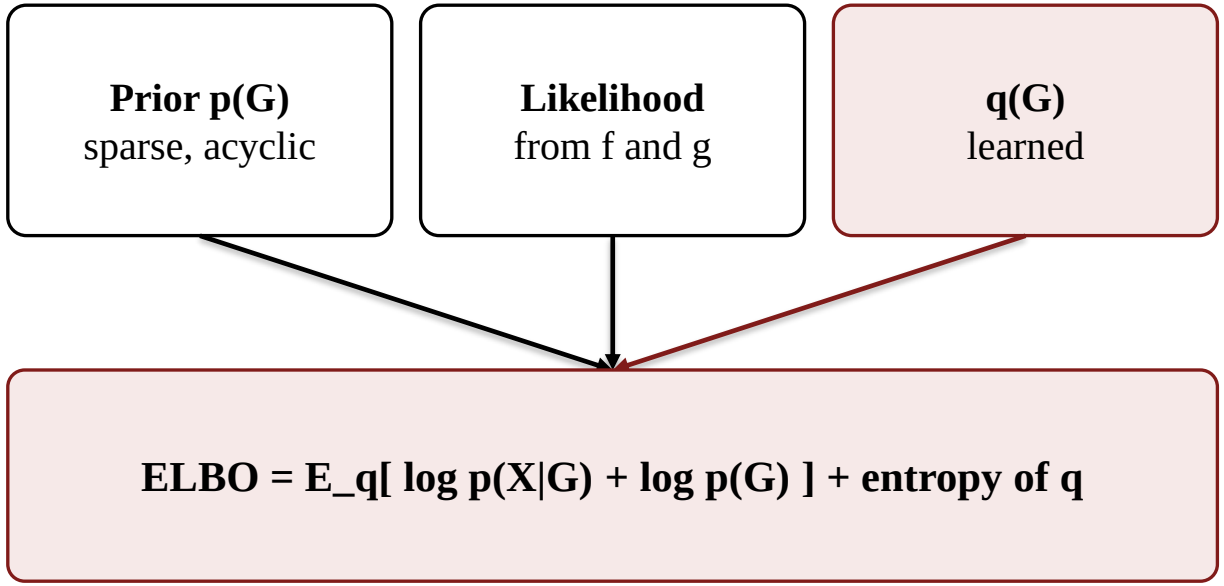


Fig. 4: The graph as one matrix per lag.

Learning A Distribution

Rhino learns a distribution over graphs.

Not one graph. It trains q(G) to explain the data while staying sparse and acyclic.



Acyclicity penalty, tightened by an augmented Lagrangian.
Gumbel-softmax gives gradients through graph samples.

Fig. 5: The training objective of Rhino.

What Rhino Outputs

Every possible edge gets a probability.

The paper ranks them (AUROC) or rounds at 0.5 (F1).

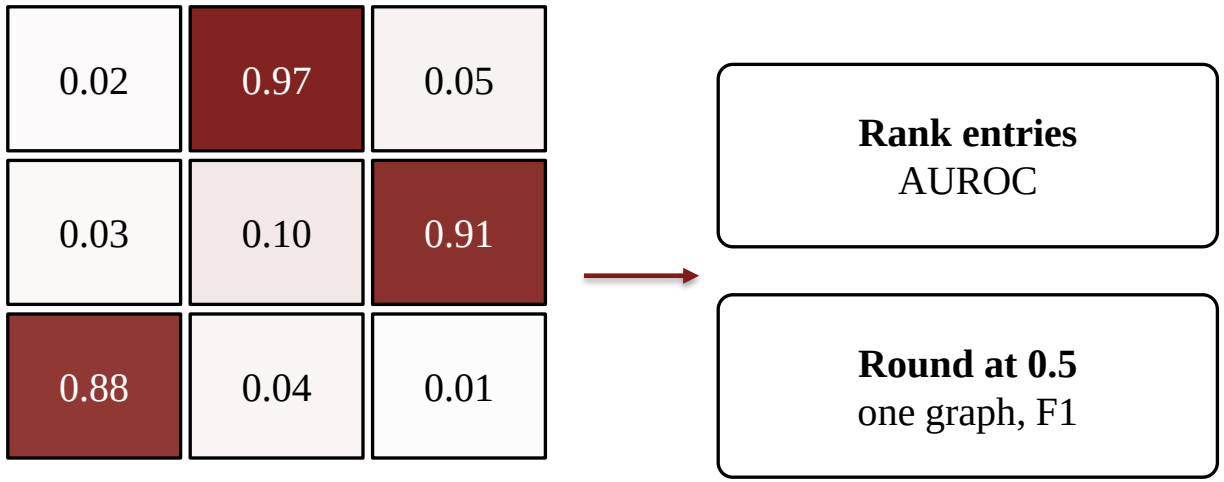
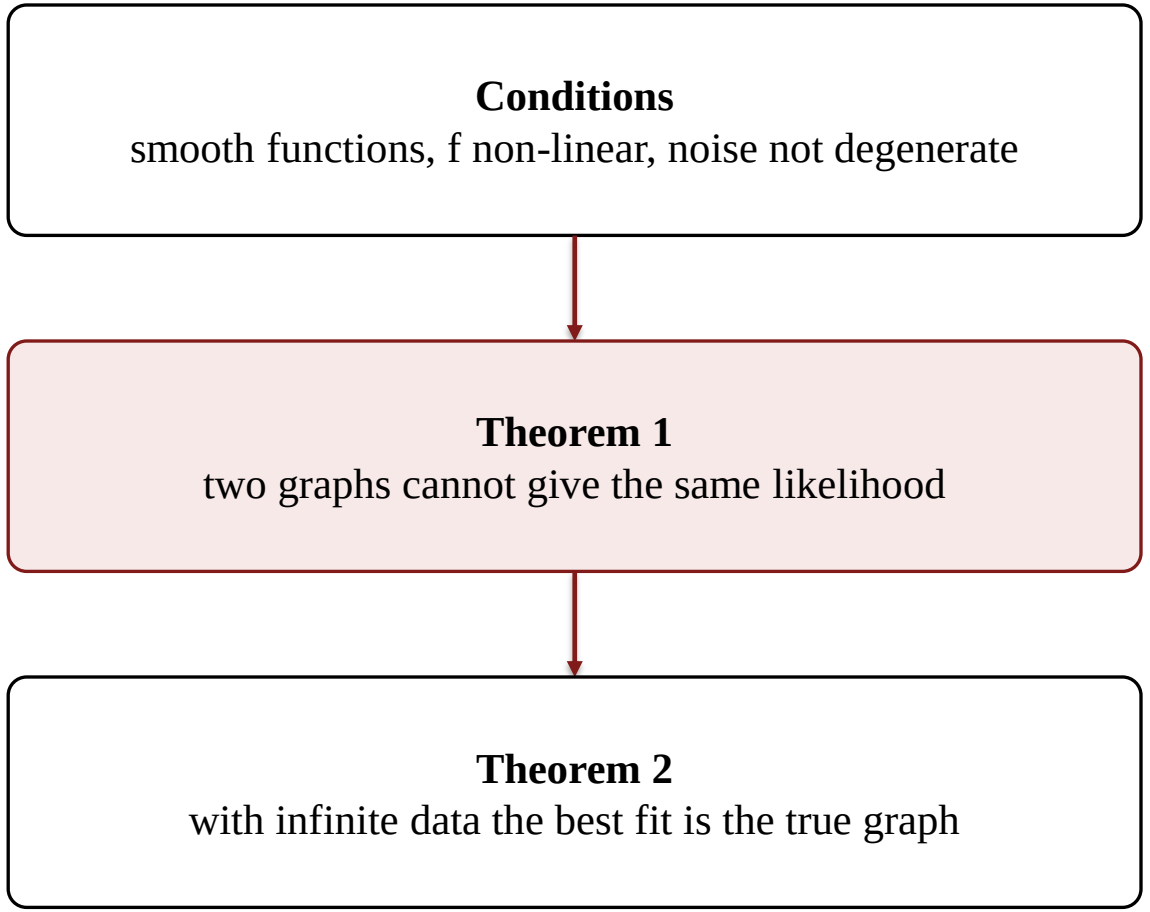


Fig. 6: Edge probabilities (illustrative values).

Why The Target Is Well Defined

In theory the true graph is unique.

The paper proves Rhino is identifiable. In practice training stops at a local optimum.



How The Paper Tests Rhino

Three benchmarks, always scored against a known graph.

F1 on synthetic data, AUROC on DREAM3 and Netsim.

Synthetic
160 datasets
5 to 40 nodes, lag 2
75% mismatched to Rhino

DREAM3 genes
5 networks, 100 genes
21 time points
AUROC

Netsim brain
fMRI, 15 regions
200 time steps
AUROC

Fig. 7: Baselines: VAR-LiNGAM, PCMCI+, DYNOTEARS, cMLP, cLSTM, TCDF, SRU, eSRU.

Paper Result 1: Synthetic Data

Best or close to best on synthetic data.

Especially in low dimension. A lag that is too long costs little.

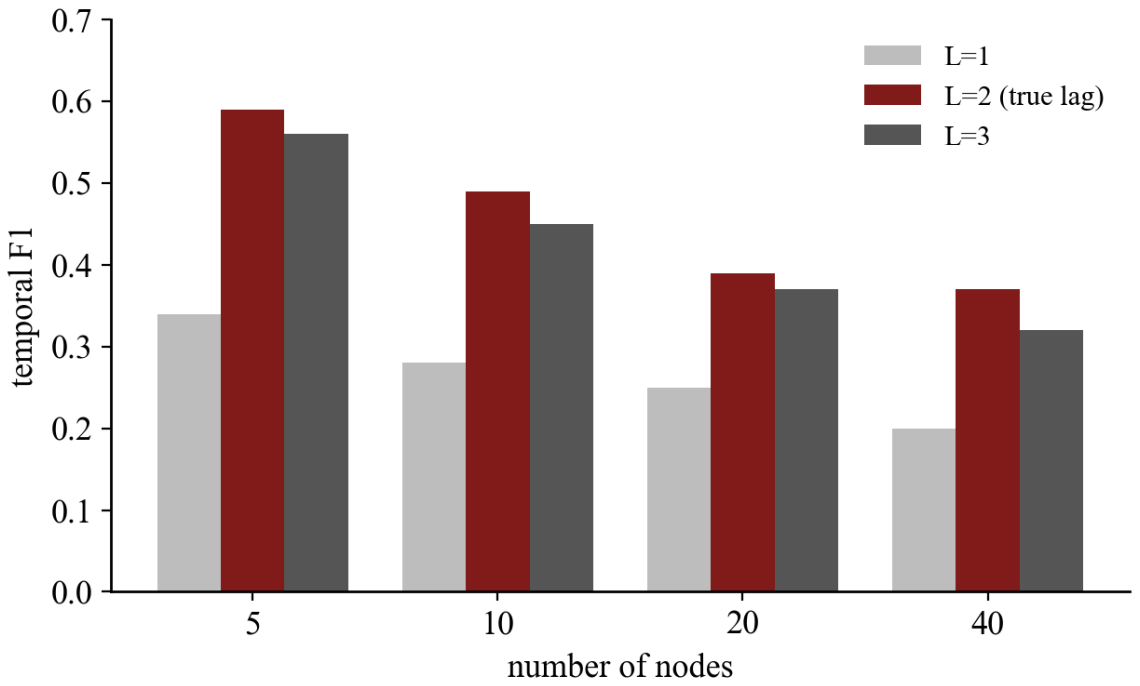


Fig. 8: Paper, Table 1: temporal F1 by model lag (true lag is 2).

Paper Result 2: DREAM3

On gene networks Rhino has the top AUROC.

Rhino or Rhino+g wins all five networks, by small margins on two.

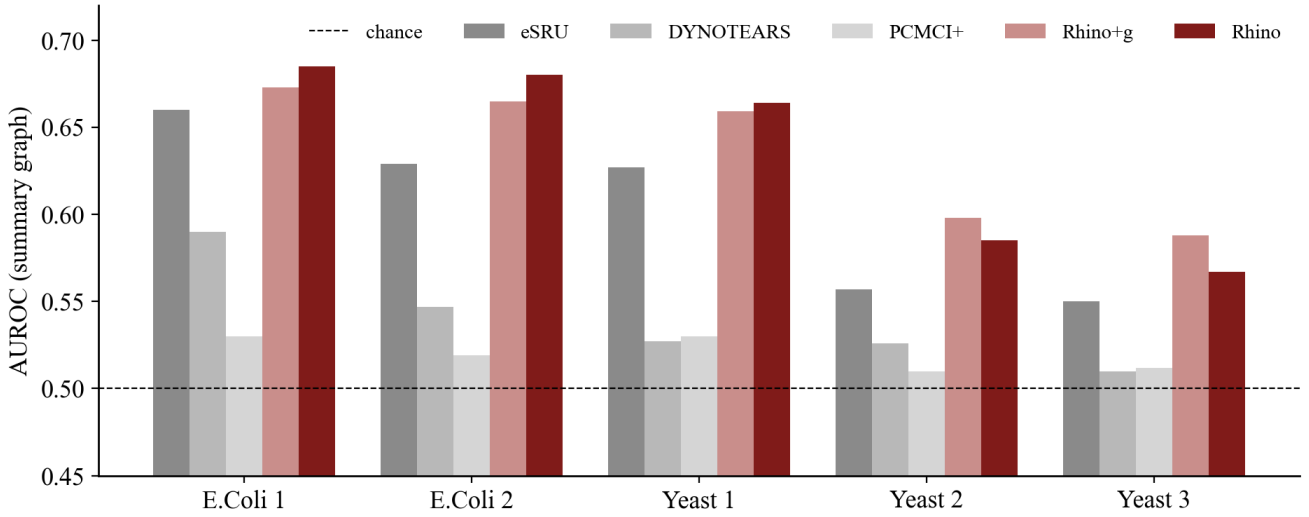


Fig. 9: Paper, Table 2: AUROC of the summary graph (subset of methods).

Paper Result 3: Netsim

On brain connectivity the AUROC is 0.99.

Without instantaneous effects it drops to 0.93.

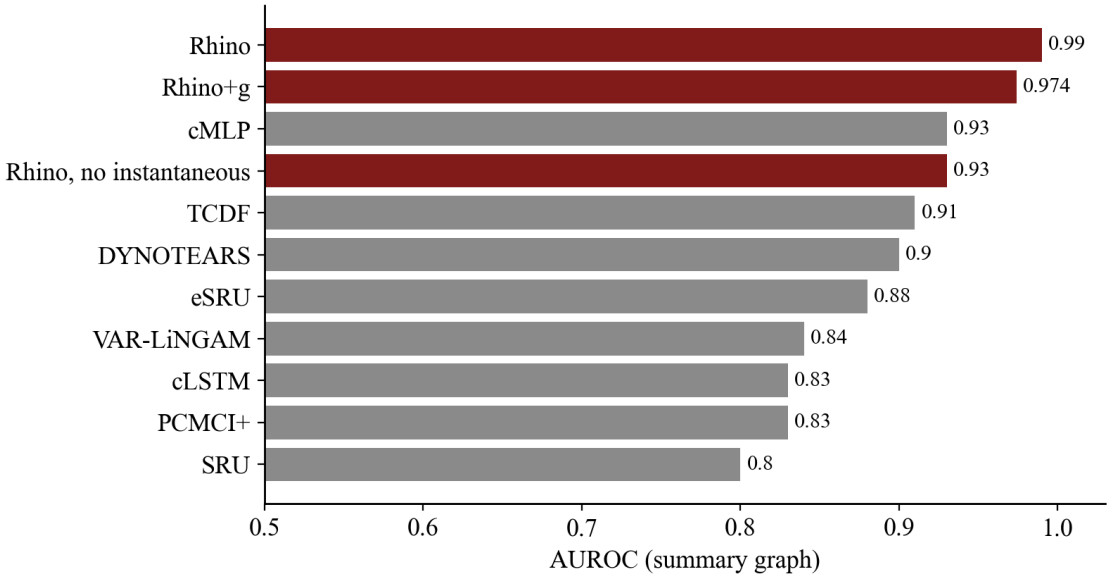


Fig. 10: Paper, Table 3: AUROC of the summary graph.

What Our M3 Run Confirmed

We reproduced the accuracy story.

Nobody tested whether the probabilities can be trusted.

Claim of the paper	M3 reproduction
History-dependent noise helps	Reproduced
Robust to a longer lag, a shorter lag hurts	Reproduced in shape
Netsim AUROC 0.99	0.987 with the paper's steps
Best or competitive on synthetic data	Partly reproduced
Rhino+s close to Rhino	Not reproduced
Probabilities are well calibrated	Never tested

What The Paper Measures

The paper checks accuracy against a known graph.

A real user has no true graph. They ask: how many of these edges are wrong?

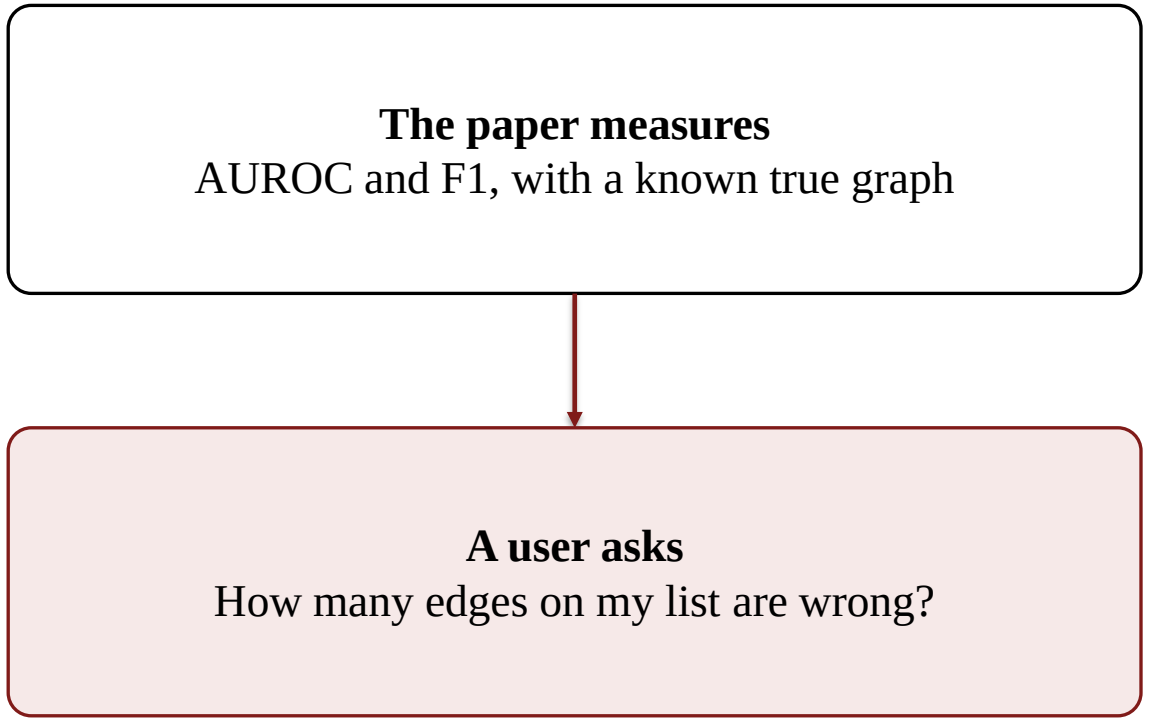


Fig. 11: Accuracy is not the same as trust.

What A Trustworthy Probability Is

A probability should match how often it is right.

Among edges scored 0.9, about 9 in 10 should be real. This is called calibration.

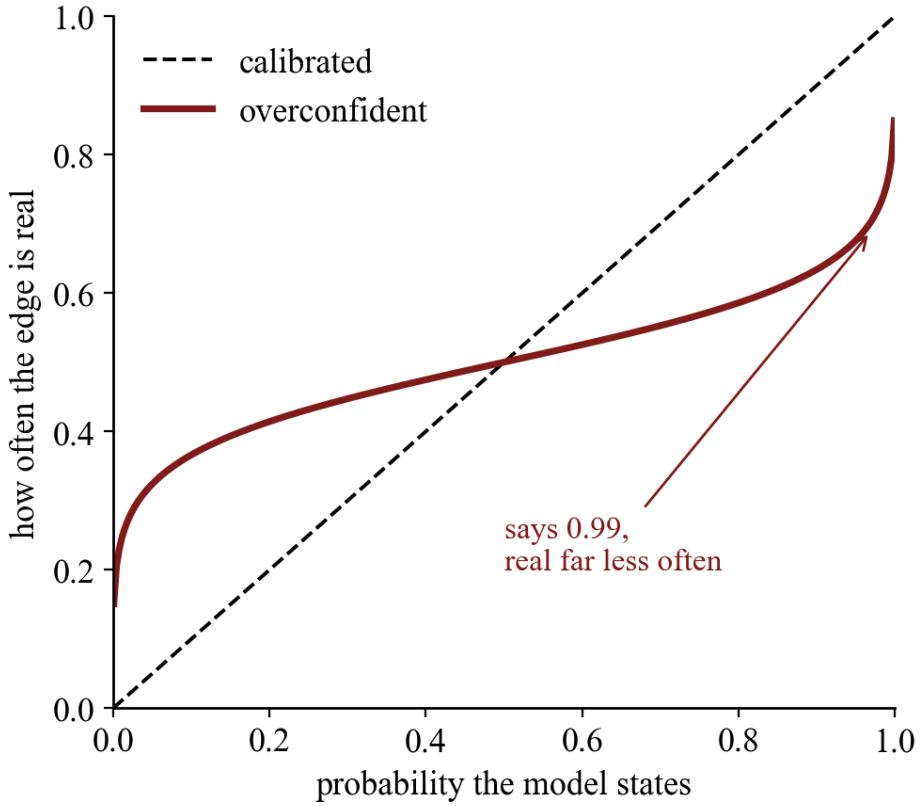


Fig. 12: Schematic, not data.

A Warning Sign

Training made Rhino more decided, not more right.

Netsim, M3 reproduction of the paper's Fig. 8, from the best step to the last.

AUROC falls
0.989 to 0.939

Validation log-likelihood falls
-31.5 to -76.1

F1 of the rounded graph rises
0.088 to 0.538

Why We Suspect The Probabilities

Four reasons to suspect them.

These are hypotheses that we test. They are not claims.

The objective rewards likelihood, not good probabilities

Independent factors ignore dependence between edges

Long training can push logits to extremes

One run on small data, never checked against truth

Our Idea

Ask for an error budget, then choose the edges.

Keep the share of wrong edges below a budget alpha, and still find as many true edges as possible.

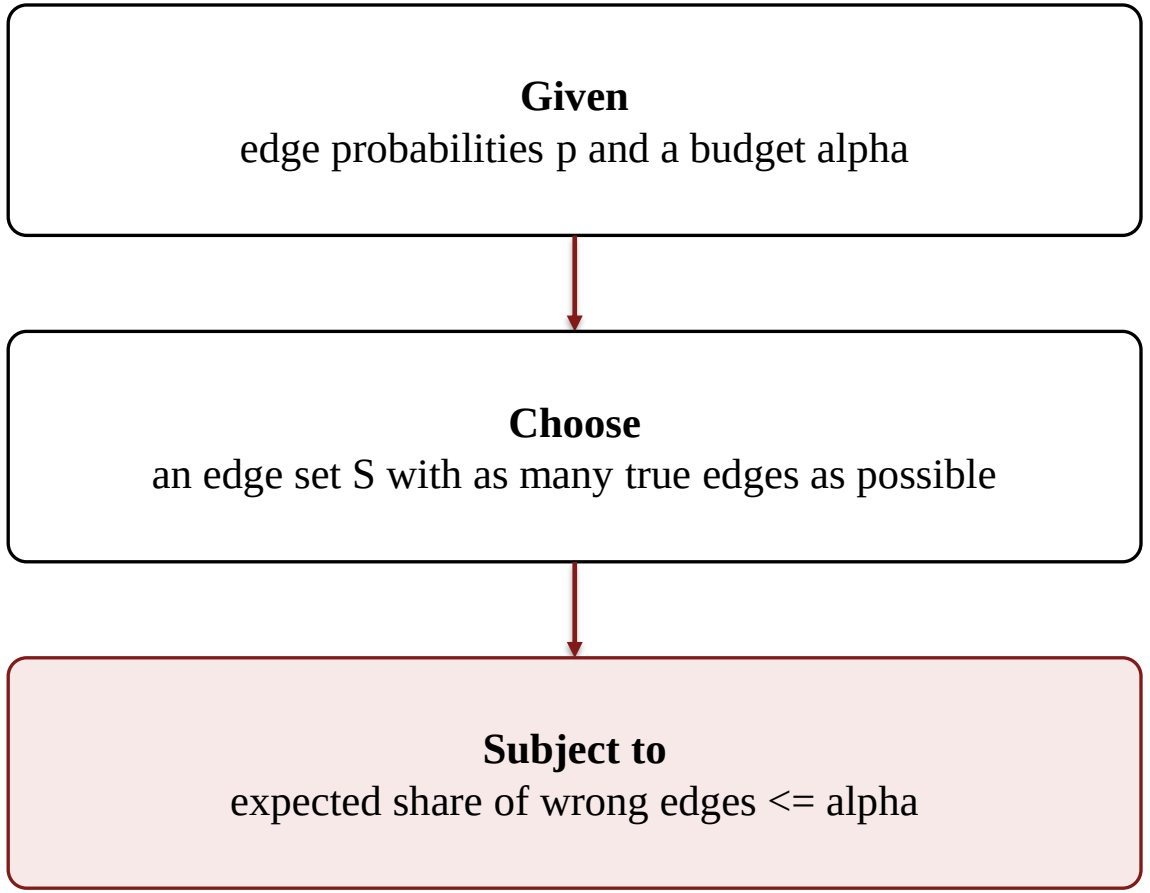


Fig. 13: Selective temporal causal discovery.

Why It Is Logical

One identity explains when a posterior fails.

Wrong share = promised error + overconfidence. If the probabilities are calibrated, the overconfidence is zero.

wrong share = promised error + overconfidence



promised error = mean of (1 - p) over the chosen edges

Fig. 14: Illustration with made-up proportions.

Why It Matters

Every edge you follow costs an experiment.

A budget turns a probability graph into a decision.

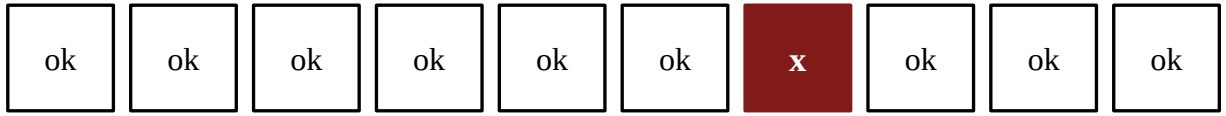


Fig. 15: Ten chosen edges, one wrong: budget 0.1.

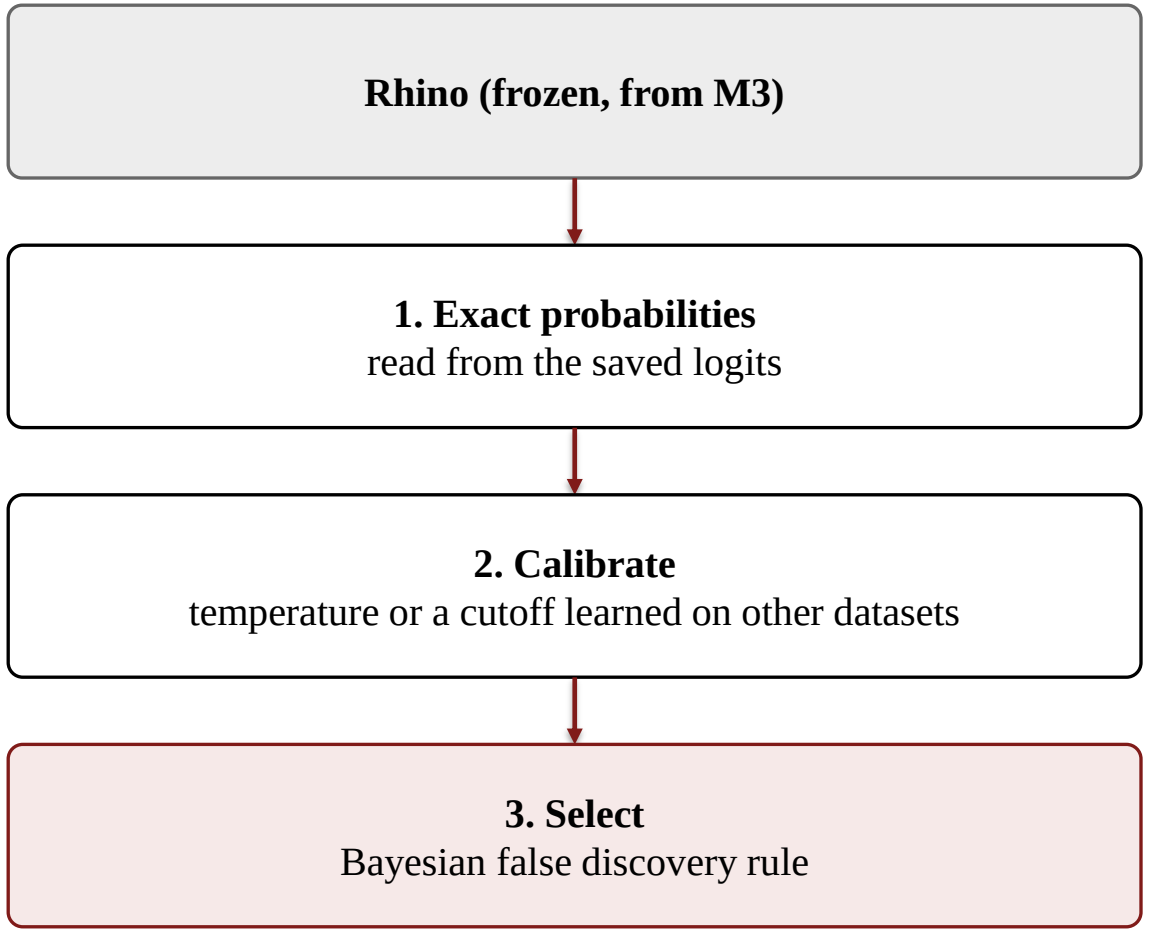
With trustworthy p
choose alpha and get the longest list within it

Without it
no way to choose how many edges to follow

Our Plan

A thin layer on a frozen Rhino.

No retraining. Labels are used only on calibration datasets.



How We Will Judge It

The key KPI is realised error against the budget.

Plus probability quality, power, and accuracy for the link to the paper.

KPI	Why
Realised share of wrong edges against budget	Is the promise true?
Brier, log-loss, calibration error	Quality of the probabilities
Power (share of true edges found)	The price of control
AUROC, F1	Link to the paper

What We Expect

Control at moderate budgets, at a price in power.

Raw Rhino should break its budget. A repair should help. Real data may not transfer.

The learned cutoff was added after we saw the first repair, so it is exploratory.

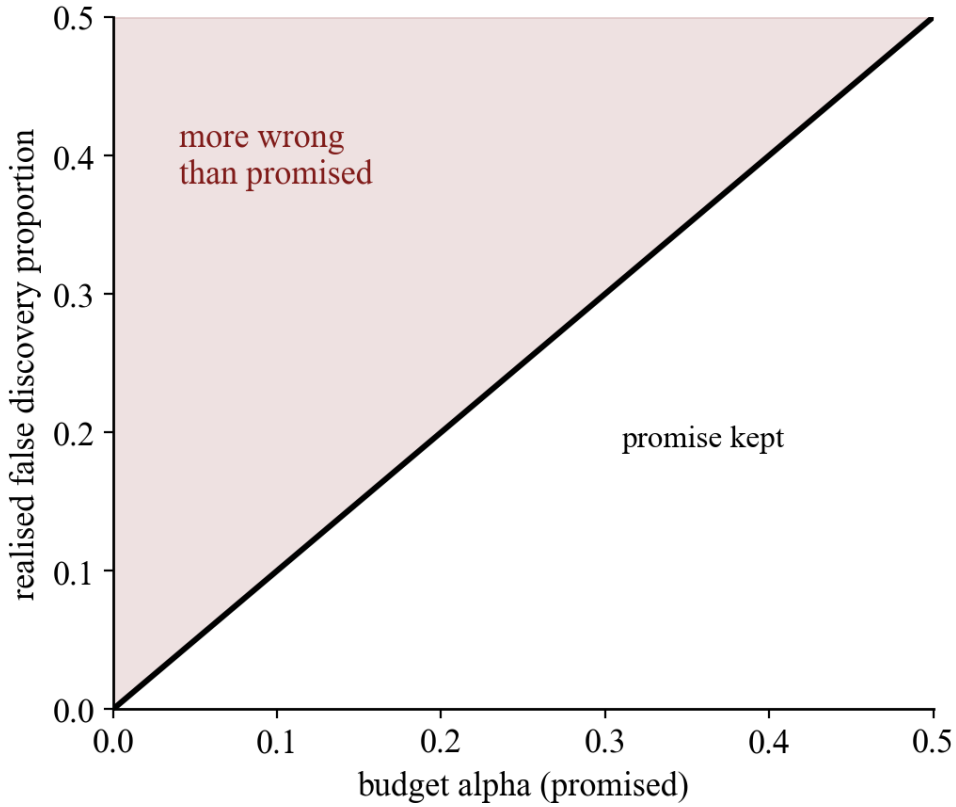


Fig. 16: Schematic of the target, not a result.

Benefits And Costs

A useful but limited tool.

We state both sides.

Benefits
A budget in the user's units
No retraining
Seconds of compute

Costs
Needs labelled graphs
Average guarantee only
Cannot fix a weak ranking
Strict budgets select nothing

Baselines And Positioning

Not a better algorithm. A trust check of Rhino.

Baselines are what the paper suggests and the standard fix.

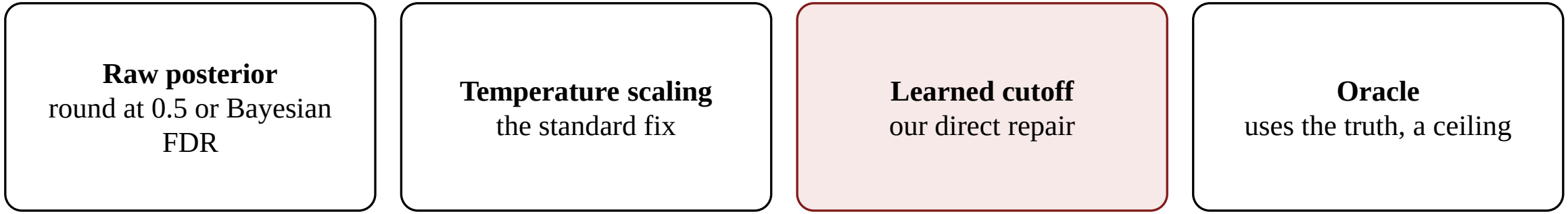


Fig. 17: Rhino (Microsoft Research, ICLR 2023) is the anchor and is not changed.

Summary

Rhino ranks edges well. Can its probabilities be trusted?

That is our question. The plan on the previous slides is how we answer it.

Base paper
strong accuracy, a probability for every edge

The gap
accuracy is measured, trust is not

Our idea
error-controlled edge selection, a thin layer on Rhino

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